PRAMAAN — Internet Protocol Detail Record Intelligence & Correlation Engine

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OFFICIAL USER MANUAL — STEP-BY-STEP FORENSIC OPERATIONS GUIDE

Gwalior Police Cyber Forensics Unit

```text
Document Version    : 2.0.0 — Full Forensic Edition
Classification      : RESTRICTED — Law Enforcement Personnel Only
Prepared By         : Pramaan IPDR Engine Development Team
Jurisdiction        : India — governed under Information Technology Act 2000, Indian Evidence Act 1872
WARNING             : Every decision made using this system may affect the liberty of a human being.
                      Operate with absolute precision. Cross-validate every finding.
```

MANDATORY FORENSIC WARNING — READ BEFORE USE

> *IPDR-based digital forensics involves analysing call data, internet session logs, and location records that belong to individual citizens protected by the Right to Privacy (K.S. Puttaswamy v. Union of India, 2017). The analytical output of this system can be used in a court of law as electronic evidence under Section 65B of the Indian Evidence Act, 1872. This means:*

1. **A single misread timestamp can falsely implicate an innocent person.** IST vs UTC confusion is one of the most common causes of wrongful accusations in digital evidence.
2. **A single misidentified IP address can destroy a case** — CGNAT (Carrier-Grade NAT) means multiple citizens share one IP address at the same time. Only the combination of IP + Port + Millisecond identifies one subscriber.
3. **A single unvalidated confidence score below 0.65 does not prove contact** — background app pings and push notifications from WhatsApp, Telegram, and other applications create false signals that mimic real calls.
4. **All evidence extracted from this system is secondary**. It must be corroborated with operator-certified records (obtained under a court order or under Section 91 CrPC / Section 92 BNSS 2023).

You are operating a forensic instrument, not a search engine. Every action taken in this system is recorded in the Audit Log with a timestamp, action type, and entity reference.

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PART A — TECHNICAL FOUNDATION

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1. WHAT IS IPDR AND WHY IT MATTERS

1.1 Definition

IPDR stands for **Internet Protocol Detail Record**. It is a log file generated automatically by a telecom operator's network infrastructure every time a mobile subscriber's device initiates or receives a data session (i.e., connects to the internet).

Think of it as a **telephone call detail record (CDR) for data traffic**. A traditional CDR records which number called which number at what time for how many seconds. An IPDR records which SIM card (MSISDN) connected to which IP address on which port using which protocol at which exact second for how long and with how many bytes transferred.

1.2 Why IPDR is Critical for Investigation

India's telecom operators are mandated under **Rule 29 of the Indian Telegraph Rules, 1951** and the **Telecom Cyber Security Rules, 2024** to retain IPDR data for a minimum of **two years** and to provide it to law enforcement agencies upon lawful request.

IPDR allows investigators to:

- Prove that suspect A and suspect B communicated over WhatsApp, Telegram, or Signal — even when both have switched off call logs on their phones.
- Identify the exact IP address, device, and location of a kidnapper or extortionist during a live call.
- Discover "burner phone" networks by tracing shared IMEI devices and overlapping session times.
- Map the complete social network of a criminal gang by analysing communication flows between MSISDNs.
- Trace cryptocurrency transaction origins by linking financial platform IP accesses to specific SIM cards.

1.3 Limitations of IPDR

- IPDR does **not** contain the content of communications. It shows metadata only (who connected where, when, for how long, how many bytes).
- IPDR for encrypted applications (WhatsApp, Signal) can prove *that* a call happened but cannot decrypt *what was said*.
- IPDR from different operators may use completely different column names, time zones, and formats, which is why this system's normalisation engine exists.

2. FULL FORENSIC GLOSSARY — EVERY TECHNICAL TERM EXPLAINED

The following glossary covers every single technical and operational term used within the Pramaan IPDR Engine. Each entry includes the full form, technical definition, and operational/forensic significance.

2.1 MSISDN

Full Form: Mobile Station International Subscriber Directory Number

Plain English: This is the complete international version of the suspect's mobile phone number.

Format Breakdown:

```text
Example: 919876543210
  91   → Country Code (CC) for India
  987  → National Destination Code (NDC), identifies operator circle
  654  → National Destination Code continued
  3210 → Subscriber Number (SN)
```

Technical Details:

- MSISDN is defined by ITU-T Recommendation E.164.
- Maximum length is 15 digits including the country code.
- Indian mobile numbers are 10 digits long. In IPDR logs, they may appear with country code (12 digits starting with 91) or without (10 digits). The system handles both.
- Some operators log the MSISDN in formats like `+919876543210`, `0091-9876543210`, or even as encrypted tokens. The ingestion engine attempts auto-detection.

Forensic Significance:

- The MSISDN is the primary forensic identifier for a suspect. It links a SIM card to a telecom subscription.
- The SIM card is registered to a person's Aadhaar/identity under TRAI regulations.
- A single MSISDN appearing on multiple IMEI devices indicates SIM swapping or SIM cloning.
- A single device (IMEI) appearing with multiple MSISDNs in a short time window indicates SIM card replacement — a hallmark of criminal tradecraft.

In the App: Every session record, extraction, PoI profile, and communication map node is indexed by MSISDN.

2.2 IMEI

Full Form: International Mobile Equipment Identity

Plain English: The unique hardware serial number burnt into every mobile phone at manufacture. Like a chassis number for a car, but for a phone.

Format Breakdown:

```
TEXT

Example: 357834901234567
  35783490 → TAC (Type Allocation Code) — first 8 digits
  123456   → SNR (Serial Number) — next 6 digits
  7        → CD (Check Digit, Luhn algorithm) — last digit
```

Technical Details:

- IMEI is a 15-digit number standardised by GSMA.
- The Luhn algorithm check digit at position 15 allows validation of the IMEI's mathematical integrity. The system validates this internally.
- IMEI is transmitted to the operator's network every time the device registers (even without a SIM inserted in some 4G scenarios).

Forensic Significance:

- The IMEI identifies the **physical handset**, independent of which SIM card is installed.
- **Shared Handset Alert:** If a single IMEI appears across multiple MSISDNs within the database, the system automatically flags this as a "Shared Handset." In criminal investigations, this pattern typically means:
  - The suspect used one phone with multiple SIM cards (SIM swapping).
  - Multiple suspects are sharing one device (gang coordination).
  - The suspect sold or discarded the phone and the new owner is also under investigation.
- In the **IMEI Intelligence** page, the system lists every MSISDN associated with each IMEI across all uploads.

In the App: Visible in Session Records, IMEI Intelligence page, and PoI Dossiers.

2.3 TAC

Full Form: Type Allocation Code

Plain English: The first 8 digits of the IMEI that identify who made the phone and what model it is.

Format Breakdown:

```
TEXT
```

```
Example: 35783490
  3578 → Reporting Body Code (who issued the TAC — e.g., BABT UK for UK-registered
devices)
  3490 → Device Model Code (specific model within that manufacturer's range)
```

Technical Details:

- TAC is managed by GSMA and officially published in the GSMA Device Registry.
- Legal authorities may request a TAC lookup through GSMA at https://www.gsma.com/services/terminal-steering-group/ to obtain the certified manufacturer and model.
- The system's TAC decoder provides model hints (e.g., "Samsung Galaxy S23") based on known TAC prefixes.

Forensic Significance:

- Knowing the device model is operationally useful. During a raid, field officers can confirm that the device found matches the model predicted by the TAC.
- Premium flagship phones (e.g., iPhone 14 Pro Max) with criminal SIM associations may indicate financial crimes (extortion money being spent).
- Cheap handsets with high session volumes may indicate a burner phone setup.

2.4 IMSI

Full Form: International Mobile Subscriber Identity

Plain English: A unique internal number embedded in the SIM card itself (distinct from the phone number on the SIM).

Format Breakdown:

```text
Example: 404201234567890
  404    → MCC (Mobile Country Code) — 404 = India
  20     → MNC (Mobile Network Code) — identifies operator (e.g., 20 = Vodafone India)
  1234567890 → MSIN (Mobile Subscriber Identification Number) — unique per subscriber
```

Technical Details:

- IMSI is stored on the SIM card's secure element.
- IMSI is transmitted over the air interface during network attachment (before TMSI is assigned).

- IMSI is different from MSISDN: IMSI is the SIM's identity; MSISDN is the user's phone number.

Forensic Significance:

- IMSI appears in some advanced IPDR formats and CDRs.
- If a suspect changes their phone number (MSISDN) but continues using the same SIM card, the IMSI remains constant — enabling tracking across number changes.
- IMSI catchers (IMSI harvesting surveillance equipment) exploit this identifier for interception, which is why it is highly sensitive.

In the App: Stored in SessionRecord if present in the source log. Visible in Session Detail View.

2.5 TMSI

Full Form: Temporary Mobile Subscriber Identity

Plain English: A short-lived, temporary code the operator's base station assigns to a mobile device to protect its IMSI from being transmitted in plain text over the air.

Technical Details:

- TMSI changes regularly (on each Location Area Update or after authentication).
- The operator maps TMSI → IMSI internally.
- TMSI appears in some operator logs and advanced IPDR exports.

Forensic Significance:

- TMSI can help link IMSI to a specific session if the mapping table is obtained from the operator.

2.6 Source IP Address

Full Form: Source Internet Protocol Address (Private / Internal)

Plain English: The IP address that the operator's internal network assigned to the suspect's device when it connected to the internet.

Technical Details:

- Based on RFC 1918 (private address ranges): `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`.
- Also based on RFC 6598 (CGNAT range): `100.64.0.0/10` (used by most Indian operators).
- This IP is **not** visible on the public internet. It is a private network address used internally within the operator's infrastructure.

Forensic Significance:

- The source IP by itself cannot be used to identify a subscriber's internet activity from external logs.
- It must be cross-referenced with the Translated IP and Translated Port to link internal subscriber activity to external internet traffic.
- Always check: if a Source IP appears as `10.x.x.x` or `100.x.x.x`, it is private/internal — it will not appear in internet traffic logs of destination servers.

In the App: Stored as `source_ip` in the SessionRecord. Displayed in session list and PoI detail view.

2.7 Source Port

Full Form: Source Port Number (Private / Ephemeral Port)

Plain English: The ephemeral (short-lived) port number assigned to a specific internet connection attempt made from the subscriber's private IP address.

Technical Details:

- Port numbers range from 0 to 65535.
- Ephemeral ports (dynamically assigned by the device's OS) typically range from 1024 to 65535.
- Each internet connection (TCP session or UDP flow) from the device gets a unique source port.
- A mobile device may open hundreds of concurrent connections, each on a different source port.

Forensic Significance:

- Along with Source IP, the Source Port forms the unique internal socket (`10.x.x.x:54231`) for one specific data connection.
- During CGNAT reverse lookup, both translated IP + translated port (the external socket) must be provided to the operator to resolve the internal source IP + source port combination — and ultimately the MSISDN.

2.8 CGNAT

Full Form: Carrier-Grade Network Address Translation

Plain English: The mechanism by which an operator translates thousands of customers' private IP addresses and ports into a single shared public IP address. Like a large apartment building where all residents appear to come from one street address to the outside world, but internally each resident has their own door number (port).

Technical Details:

- Defined in RFC 6888.
- Indian operators (Jio, Airtel, Vi, BSNL) universally deploy CGNAT.

- At the CGNAT device, the mapping is: `[Private IP : Private Port] ↔ [Public IP : Public Port]`.
- This mapping has an associated **start time** and **end time** (the NAT binding lifetime).

Forensic Significance:

- **This is the most critical concept in IPDR investigations.** If a WhatsApp server logs that IP `103.74.19.155` sent a message at `2024-03-15 14:23:45.123 IST`, you cannot identify who sent it using only the IP. You need:
  1. The exact public IP: `103.74.19.155`
  2. The exact public port used: e.g., `54231`
  3. The exact timestamp (millisecond-accurate): `2024-03-15 14:23:45.123 IST`
     Then, the operator's CGNAT logs will return the MSISDN that was assigned this translation at that moment.
- Obtaining CGNAT logs requires a separate legal process (court order or 91-BNSS notice) from the IP address owner (operator).

2.9 Translated IP (NAT Public IP)

Full Form: NAT-Translated (Network Address Translated) Public IP Address

Plain English: The public IP address visible to the internet — the one that external servers (like WhatsApp, Telegram, Google) see when the suspect's device connects to them.

Technical Details:

- This is a globally routable public IP address assigned to the operator's CGNAT device.
- It belongs to the operator's IP space (ASN).
- It changes as subscribers' NAT bindings expire and are reassigned.

Forensic Significance:

- This IP appears in server-side logs of internet platforms. If obtained from a MLAT (Mutual Legal Assistance Treaty) request or from the company's law enforcement response team, this IP + translated port + timestamp can be reverse-matched to the suspect's MSISDN.
- This field is the bridge between internal carrier records and external internet platform evidence.

In the App: Stored as `translated_ip`. Displayed alongside `translated_port` in Session Detail View.

2.10 Translated Port (NAT Public Port)

Full Form: NAT-Translated (Network Address Translated) Public Port Number

Plain English: The specific public port number that the CGNAT device assigned to represent this subscriber's specific connection to the outside world.

Technical Details:

- Every subscriber-initiated connection gets a unique `[public IP : public port]` pair called a NAT binding.
- Port range 1024–65535 is typically used for NAT bindings.
- Multiple subscribers may share the same public IP but each will have a different translated port number.

Forensic Significance:

- The translated port is the granular key that uniquely identifies one subscriber's specific session among thousands sharing one public IP at any given instant.
- In any external platform MLAT response, always request translated port alongside the timestamp. Without it, the public IP alone is useless for CGNAT identification.

2.11 Destination IP Address

Full Form: Destination Internet Protocol Address

Plain English: The IP address of the external server or other device that the suspect's phone connected to.

Technical Details:

- This is a public IP address (or private IP in LAN-level sessions) of the counterpart of the connection.
- Destination IPs are classified by the system based on known IP ownership (ASN lookup) and custom platform ranges configured in Settings.

Forensic Significance:

- Known platform IPs (e.g., `157.240.x.x` = Meta/WhatsApp, `149.154.x.x` = Telegram) indicate which application the suspect was using.
- Unknown IPs may indicate:
  - A VPN server (masking true destination).
  - A Peer-to-Peer media stream (direct call with another suspect).
  - A criminal web server, dark web proxy, or C2 (Command & Control) server.
- The system cross-references destination IPs against its Platform Ranges database to auto-classify them.

2.12 Destination Port

Full Form: Destination Port Number

Plain English: The specific service port on the destination server that the suspect's device was connecting to.

Technical Details:

- Ports range from 0 to 65535.
- Well-known ports (0–1023) are assigned by IANA to specific services.
- Common ports in IPDR evidence:

```text
Port 80   → HTTP (unencrypted web)
Port 443  → HTTPS (encrypted web, WhatsApp, Telegram text, etc.)
Port 5222 → XMPP (used by some messenger apps)
Port 3478 → STUN (Session Traversal Utilities for NAT — used for VoIP call setup)
Port 3479 → STUN alternate
Port 5349 → STUN over TLS
Port 19302 → Google STUN (WebRTC / Google Meet)
Port 443 UDP → QUIC protocol (YouTube, Google, WhatsApp calls)
Port 50318 → WhatsApp voice (varies)
Dynamic ports 10000–65535 → Used for RTP (Real-Time Protocol) media streams in
VoIP calls
```

- Ports 3478, 3479, 3480, and 5349 appearing in UDP sessions of short duration indicate VoIP call **signalling** (the handshake before the call).
- UDP sessions on high ephemeral ports (>10000) lasting more than 10 seconds with both upload and download traffic indicate an active voice or video call media stream (RTP/SRTP).

Forensic Significance:

- Port 443 TCP traffic = web activity or encrypted app messaging (cannot distinguish between types).
- Port 443 UDP traffic = likely QUIC protocol (WhatsApp calls use QUIC, as does Google).
- Port 3478 UDP traffic = STUN signalling (call setup confirmation, but not the actual call audio stream).
- High-ephemeral port UDP > 10s duration = almost certainly a VoIP call media stream.

2.13 Protocol

Full Form: Internet Protocol Transport Protocol

Plain English: The type of "language" (connection method) used for the internet connection. The two most common are TCP and UDP.

Technical Details:

- **TCP (Transmission Control Protocol):** Connection-oriented, guarantees delivery, has acknowledgement mechanisms. Used for: web browsing (HTTP/S), file downloads, text messaging, email.
- **UDP (User Datagram Protocol):** Connectionless, no delivery guarantee, very fast. Used for: voice calls (VoIP), video calls, live streaming, gaming, DNS queries.
- **ICMP (Internet Control Message Protocol):** Used for network diagnostics (ping). Rarely appears in IPDR.

Forensic Significance:

- A TCP session to port 443 indicates web/app activity.
- A UDP session to a high port lasting >10 seconds with bidirectional byte counts indicates a voice/video call — this is the highest-value P2P call indicator.
- Mixed TCP+UDP sessions to the same destination IP at nearly the same time indicate: text messaging (TCP 443) followed by a voice call (UDP) on the same application.

2.14 Classification

Full Form: Session Traffic Classification

Plain English: The category the system assigns to each internet session based on automatic analysis of destination IP, port number, protocol, duration, and byte count.

Possible Values:

CLASSIFICATION	WHAT IT MEANS	FORENSIC VALUE
`p2p`	Peer-to-Peer: Direct device-to-device media stream, likely an active voice or video call	**Highest** — Proves direct communication between two devices
`relay`	Server-relayed: Traffic through a known application server (WhatsApp relay, Telegram relay)	**Medium** — Proves use of application but not who the counterpart is
`unknown`	Cannot be confidently classified	**Low** — May be background traffic, push notification, VPN, or unknown service

How Classification Is Determined:

1 The system looks up the destination IP against custom Platform Ranges (Settings > Platform Ranges).

2. If the IP is a known platform relay (e.g., Meta/WhatsApp CIDR block), the session is classified `relay` with confidence 0.85.
3. If the destination IP is NOT a known platform AND the session uses UDP AND port is >= 1024 AND duration > 10 seconds AND both bytes_up and bytes_down > 0, the session is classified `p2p` with confidence 0.90.
4. STUN/TURN ports (3478, 3479, 3480, 5349) are classified as `relay` with confidence 0.70.
5. Everything else is `unknown`.

2.15 Confidence

Full Form: Classification Confidence Score

Plain English: A number between 0.00 and 1.00 (0% to 100%) expressing how certain the system is about its classification of this session.

Score Breakdown:

```text
0.00 – 0.49  → Very low confidence. Do NOT use as evidence.
0.50 – 0.64  → Low confidence. Background noise, pings, push notifications.
0.65 – 0.79  → Moderate confidence. Corroborative but not standalone evidence.
0.80 – 0.89  → High confidence. Strong indicator of actual communication event.
0.90 – 1.00  → Very high confidence. Near-certain classification. Suitable as primary supporting evidence.
```

Forensic Significance:

- The system default extraction threshold is 0.65. Do not lower this below 0.50 under any circumstances when building court submissions.
- A p2p session with confidence 0.90 and duration 120 seconds is strong forensic evidence of a 2-minute direct VoIP call.
- A relay session with confidence 0.85 is strong evidence the suspect was using WhatsApp at that moment.
- NEVER use confidence 0.48 (unknown) sessions as evidence of communication.

2.16 PoI

Full Form: Person of Interest

Plain English: A specific suspect or target individual in the investigation, identified by their MSISDN (phone number).

What the PoI Profile Contains:

- Total sessions (all internet activity logs ingested for this MSISDN).
- P2P session count (potential direct calls).
- Relay session count (encrypted app usage).
- First seen and last seen timestamps in the evidence database.
- Total bytes transferred (indicates data volume, useful for profiling communication intensity).
- Associated IMEIs (physical devices used).
- Top locations visited (base station Cell IDs).
- Top applications used (by session count and duration).
- Top destination IPs contacted.
- WhatsApp-specific interaction log.

Forensic Significance:

- The PoI Dossier is the primary intelligence product generated by this system and is the document most likely to be used in court.
- It must be exported as PDF and certified by the IO (Investigating Officer) before submission.

2.17 B-Party

Full Form: B-Party = the Second Party (Counterpart) in a Communication

Plain English: In Indian law enforcement terminology, the "A-Party" is the subscriber whose records you have obtained (the primary target). The "B-Party" is the person they are communicating with.

Technical Details:

- In a direct P2P VoIP call, the A-Party's IPDR shows: `source_ip:source_port → destination_ip:destination_port` using UDP.
- The destination_ip:destination_port is the B-Party's public NAT address.
- To identify the B-Party, investigators must approach the operator serving that destination IP and request CGNAT reverse lookup using those coordinates.
- The Pramaan extraction engine automates the generation of **Request Packages** — formatted legal notices pre-filled with these coordinates for each operator.

Forensic Significance:

- B-Party identification is the primary objective of most IPDR analysis investigations.

- Each candidate B-Party IP address extracted from the A-Party's IPDR must be sent to the relevant operator as a CGNAT search request with exact timestamp + port number.
- The operator's response will confirm or deny the MSISDN behind that IP at that moment.

2.18 ASN

Full Form: Autonomous System Number

Plain English: A unique number that identifies a specific block of IP addresses owned and operated by one organisation (e.g., a company, a university, a telecom operator, or a technology platform).

Examples:

```text
AS32934  → Meta Platforms Inc. (Facebook, WhatsApp, Instagram)
AS15169  → Google LLC (YouTube, Google Meet, Gmail)
AS62041  → Telegram Messenger Inc.
AS396903 → Signal Messenger LLC
AS714    → Apple Inc. (iMessage, FaceTime)
AS8075   → Microsoft Corporation (Teams, Skype, OneDrive)
AS6432   → Discord Inc.
AS55836  → Reliance Jio Infocomm Limited
AS9498   → Bharti Airtel Limited
AS55410  → Vodafone Idea Limited
AS45820  → BSNL India
```

Forensic Significance:

- When a destination IP resolves to AS32934 (Meta), you can confirm the subscriber was accessing a Meta-owned server (WhatsApp or Facebook).
- When a destination IP resolves to AS15169 (Google), it could be YouTube, Gmail, Drive, or Google Meet — context (port, duration, bytes) distinguishes them.
- Unknown ASNs may indicate VPN operators, offshore hosting, or dark web infrastructure.

2.19 CIDR

Full Form: Classless Inter-Domain Routing

Plain English: A compact notation for specifying a range of IP addresses. Used in Platform Ranges settings to define which IP blocks belong to which app.

Format Breakdown:

```text
Example: 157.240.0.0/16
  157.240.0.0  → The starting IP address of the range
  /16          → Subnet mask — means the first 16 bits are fixed (the "network" part)
                 This gives 2^(32-16) = 65,536 IP addresses in this block
                 Range: 157.240.0.0 → 157.240.255.255
```

Forensic Significance:

- When a suspect's session shows a destination IP of `157.240.12.5`, the system checks it against the CIDR `157.240.0.0/16` (Meta/WhatsApp range) and classifies the session accordingly.
- Keeping platform CIDR ranges up-to-date in Settings is critical for accurate classification. Technology companies regularly update their IP blocks.

2.20 Cell ID

Full Form: Cell Global Identity (CGI) / E-UTRAN Cell Global Identifier (ECGI)

Plain English: A unique numerical code identifying the specific base transceiver station (cell tower) sector that a mobile device was connected to when the session was recorded.

Format Breakdown:

```text
2G/3G CGI: MCC-MNC-LAC-CI
  Example: 404-20-5310-12345
  404   → MCC (India)
  20    → MNC (Vodafone India)
  5310  → LAC (Location Area Code)
  12345 → CI (Cell Identity number)

4G ECGI: MCC-MNC-eNB_ID-CI
  Example: 404-20-123456-01
  eNB_ID → eNodeB ID (the 4G base station number)
  CI     → Cell Index within that eNodeB
```

Forensic Significance:

- The Cell ID provides **approximate geolocation** of the suspect at the time of the session. A cell tower sector typically covers 200 metres to 5 kilometres depending on whether it is urban or rural.
- Cell ID data in IPDR confirms the suspect was physically present in a specific area at the time of the session.

- The system uses Cell ID data to generate the Location Summary and Top Locations section of the PoI Dossier.
- **Important:** Cell ID only gives the tower location, not the device's exact GPS coordinates. For precise location, a separate CDR + tower coordinates lookup is required.

2.21 RTP / SRTP

Full Form: Real-Time Transport Protocol / Secure Real-Time Transport Protocol

Plain English: The actual protocol that carries voice and video audio/visual data during a VoIP call.

Technical Details:

- RTP runs over UDP.
- SRTP is the encrypted version used by WhatsApp, Signal, and FaceTime.
- RTP/SRTP sessions on high ephemeral ports (usually 10000–65535) are the forensic signature of an active voice or video call.
- Duration of the RTP stream = duration of the call.

Forensic Significance:

- When a UDP session is observed between a suspect's MSISDN and an unknown external IP on a high dynamic port lasting >10 seconds with both up and down bytes, the system classifies this as `p2p` — indicating a real-time media stream (call).

2.22 STUN / TURN

Full Form: Session Traversal Utilities for NAT / Traversal Using Relay NAT

Plain English: STUN and TURN are protocols used by VoIP apps (WhatsApp, Signal, Zoom, Teams) to establish call sessions through NAT (CGNAT) barriers before the actual call audio starts.

Technical Details:

- STUN (Port 3478) is the call setup negotiation handshake. It helps two devices discover each other's public NAT IP addresses.
- TURN (Port 3478 or 443) is used when a direct P2P path cannot be established — the audio is relayed through a server.
- If you see a STUN UDP session at port 3478, it means a VoIP call was *initiated* at that timestamp. The call audio will follow on a different dynamic port immediately after.

Forensic Significance:

- A STUN session (port 3478) at time T followed by a UDP stream on a dynamic port at time T+1 is the forensic signature of an attempted and then connected VoIP call.
- The STUN session alone does not prove the call succeeded — the subsequent dynamic UDP stream confirms it.

2.23 VoIP

Full Form: Voice over Internet Protocol

Plain English: Any voice call made over the internet rather than the traditional telephone network. Includes WhatsApp calls, Signal calls, Telegram voice messages, Google Meet, Zoom, Microsoft Teams, FaceTime, and Skype.

Forensic Significance:

- All VoIP calls leave traces in IPDR data.
- Identifying a VoIP call in IPDR requires: UDP protocol + high destination port (>1024) + session duration >10 seconds + bidirectional byte transfer.
- WhatsApp VoIP calls to its relay infrastructure will show as `relay` classification (you can prove the suspect was on a WhatsApp call but cannot identify the B-party from the IPDR alone).
- Direct P2P VoIP sessions (when two suspects' phones connect directly) will show as `p2p` classification — and the destination IP can be reverse-CGNAT-searched to identify the B-party.

2.24 Bytes Up / Bytes Down

Full Form: Bytes Uploaded (Uplink) / Bytes Downloaded (Downlink)

Plain English: The volume of data sent from the suspect's device to the internet (bytes_up) and received from the internet by the suspect's device (bytes_down).

Forensic Significance:

- Symmetrical (roughly equal) bytes_up and bytes_down during a UDP session strongly indicates a two-way voice call (both parties talking and listening simultaneously).
- Asymmetrical bytes (e.g., very high bytes_down, near-zero bytes_up) indicates one-way data consumption: downloading a file, streaming a video, receiving a large image.
- Very low byte counts (< 500 bytes) in a session likely indicate a push notification ping, heartbeat check, or DNS query — NOT an active communication event.
- High byte counts (>1MB in 30 seconds) with UDP protocol indicate audio/video streaming.

2.25 Duration (Session Duration Seconds)

Full Form: Session Duration in Seconds

Plain English: How many seconds the internet session lasted from connection establishment to disconnection.

Forensic Significance:

- Duration < 5 seconds: App health check, push notification, DNS query. Extremely low forensic value.
- Duration 5–10 seconds: Possible message delivery confirmation. Low forensic value.
- Duration > 10 seconds with UDP: Near-certain voice/video call. High forensic value.
- Duration > 60 seconds with UDP: Definitive ongoing voice/video call. Highest forensic value.
- Duration = 0 with UDP: Single-packet event (e.g., SIP invite or STUN binding request).

2.26 IP Allocation Type

Full Form: IP Address Allocation Type

Plain English: Whether the operator assigned the subscriber a static IP (always the same) or a dynamic IP (changes every session).

Possible Values: `static`, `dynamic`, `CGNAT`, `unknown`

Forensic Significance:

- Dynamic IP allocation (most common in India): The subscriber gets a new private IP each time they reconnect to the network. CGNAT translation is performed on top.
- Static IP allocation (rare, mostly for enterprise connections): The subscriber always gets the same IP address. Much easier to identify in external logs.
- This field in carrier logs is often labelled `static_dynamic_ip_address_allocation` (DoT standard format) or `allocation_type`.

2.27 Domain

Full Form: Domain Name (Fully Qualified Domain Name, FQDN)

Plain English: The human-readable website or server name extracted from DNS queries or connection metadata.

Examples: `api.whatsapp.com`, `t.me`, `signal.org`

Forensic Significance:

- Some operator IPDR formats include the resolved domain name alongside the IP address.
- If a domain name is present, it provides clear, human-readable evidence of which service the suspect accessed.
- Domain evidence is typically more court-friendly than raw IP addresses.

2.28 Record Type

Full Form: IPDR Record Type

Plain English: The category of log record — whether it is a full IPDR session record or a NAT translation log.

Possible Values:

- `ipdr`: Standard IPDR record with source IP, destination IP, and usage data.
- `nat_syslog`: A raw CGNAT translation log (contains public IP/port → private IP/port mappings, typically without session application data).
- `ipdr_nat`: A combined record that includes both the session data and the CGNAT translation in one row (the ideal format, less common).

2.29 Import Specification

Full Form: Custom Column Mapping Import Specification

Plain English: A user-defined template that tells the system how to interpret a non-standard IPDR log format by mapping the log's column names to the system's canonical field names.

When Needed:

- When the operator's log uses non-standard column names (e.g., `calling_party_number` instead of `msisdn`).
- When the operator combines IP and port into a single column (e.g., `src_ip_port` = `10.5.3.2:54312`).
- When the log uses a non-standard date format.

How It Works:

1. Create the spec in Settings > Import Specs.
2. Give it a name (e.g., `Jio Circle - Bhopal IPDR Format`).
3. Map each source column to the target canonical field.
4. Save and select this spec when uploading.

2.30 Adapter

Full Form: Operator Format Adapter Profile

Plain English: A built-in detection profile that automatically recognises standard IPDR formats from major Indian operators without requiring a custom mapping.

Built-in Adapter Profiles:

ADAPTER NAME	REQUIRED COLUMNS
DoT IPDR (standard)	`landline_msisdn_mdn_leased_circuit_id`, `source_ip_address`, `source_port`, `destination_ip`, `destination_port`, `static_dynamic_ip_address_allocation`
NAT SYSLOG	`start_date_time`, `source_ip_address`, `source_port`, `translated_ip_address`, `translated_port`, `destination_ip_address`, `destination_port`
Airtel	`a_party_msisdn`, `source_ip`, `source_port`, `dest_ip`, `dest_port`, `session_duration`, `uplink_bytes`, `downlink_bytes`
Jio	`msisdn`, `src_ip`, `src_port`, `translated_ip`, `translated_port`, `destination_ip`, `destination_port`
Vodafone Idea	`subscriber`, `private_ip`, `private_port`, `remote_ip`, `remote_port`, `tx_bytes`, `rx_bytes`
BSNL	`mobile_number`, `allocated_ip`, `source_port`, `server_ip`, `server_port`, `duration_sec`
Generic Canonical	`msisdn`, `source_ip`, `source_port`, `destination_ip`, `destination_port`

2.31 Quarantine

Full Form: Quarantined (Rejected) Row Record

Plain English: A data row from the uploaded file that the system could not process because it failed validation checks.

Common Quarantine Reasons:

- Invalid IP address format (e.g., `999.0.0.1` — not a valid IP).
- Invalid MSISDN (empty, too short, or contains letters).
- Unparseable timestamp (date in unrecognised format).
- Missing required fields (destination_ip or destination_port absent).
- Port number out of valid range (not between 0–65535).

Forensic Significance:

- Quarantined rows should be reviewed carefully. If a large number of valid-looking rows are quarantined, the file may have a non-standard format requiring a custom Import Specification.

- Quarantine rates above 30% of total rows should trigger a format investigation before proceeding with analysis.

2.32 Platform Ranges

Full Form: Application Platform IP CIDR Ranges

Plain English: A database of known IP address blocks belonging to major communication apps (WhatsApp, Telegram, Signal, etc.). The system uses this to automatically classify which sessions belong to which application.

Built-in Platform Ranges (Default):

PLATFORM	CIDR	ASN
Meta/WhatsApp	157.240.0.0/16	AS32934
Meta/WhatsApp	69.171.224.0/19	AS32934
Meta/WhatsApp	31.13.64.0/18	AS32934
Telegram Messenger	149.154.160.0/20	AS62041
Telegram Messenger	91.108.4.0/22	AS59930
Signal Messenger	142.251.209.0/24	AS396903
Google Duo/Meet	74.125.0.0/16	AS15169
Apple iMessage	17.0.0.0/8	AS714
Discord VoIP	66.22.196.0/22	AS6432
Microsoft Teams	52.112.0.0/14	AS8075

2.33 Suspicious Patterns

Full Form: Auto-detected Suspicious Behavioural Patterns

Plain English: Automated anomaly detections that the system's analytics engine flags when it identifies statistically unusual behaviour in the evidence database.

Pattern Types and Their Severity:

PATTERN	SEVERITY	DESCRIPTION
Burst Activity	Medium	MSISDN with unusually high session count in a short time window

PATTERN	SEVERITY	DESCRIPTION
Shared Handset	High	Single IMEI used by multiple MSISDNs
Night Activity	Medium	MSISDN primarily active between 00:00–05:00 IST
VPN Usage	High	Sessions destined for known VPN provider IP ranges
Darknet Traffic	High	Sessions to known Tor exit node IPs
Cross-Operator P2P	High	P2P sessions between MSISDNs on different telecom operators
Isolated Node	Low	MSISDN with only one contact in the communication graph

2.34 Request Package

Full Form: CGNAT Search Request Package (Legal Notice)

Plain English: An automatically generated, formatted document containing the exact IP address, port number, and timestamp required for an operator to perform a CGNAT reverse lookup and identify the B-party MSISDN.

Contents of a Request Package:

- Target Operator (ASN-identified, e.g., "Reliance Jio Infocomm Ltd.")
- Session timestamp (IST, millisecond-accurate)
- Translated (public) IP address
- Translated (public) port number
- Case reference number
- IO (Investigating Officer) name and designation

Forensic Significance:

- These packages are the direct output that investigators send (via legal channels — court order, Section 91 CrPC notice) to telecom operators to de-anonymise B-party suspects.
- The correctness of these packages is mission-critical. A single wrong digit in the port number will return no match from the operator.

2.35 Audit Log

Full Form: System Audit Trail

Plain English: A chronological, tamper-evident record of every action performed in the system: file uploads, extraction runs, case creations, report generations, and factory resets.

Fields Recorded:

- Timestamp (IST)
- Action type (e.g., `upload_ingested`, `extraction_created`, `reset_executed`)
- Entity type (e.g., `upload`, `case`, `extraction`)
- Entity ID (e.g., `UPL-4A3B7C`)
- User (defaults to `system` in single-user mode)
- IP address of client

Forensic Significance:

- The audit log establishes chain of custody for digital evidence.
- In court, you may be asked to demonstrate that the evidence was not modified after ingestion. The audit log shows when data was ingested and when reports were generated.
- Do not clear the audit log during an active investigation.

---

3. THE DATA FLOW: FROM TELECOM TOWER TO THIS DASHBOARD

Understanding this flow is essential to correctly interpreting the evidence:

```text
[Mobile Device] → connects to internet
     ↓
[Operator Base Station] (Cell Tower) → assigns Cell ID
     ↓
[Operator's Private Network]
     ↓ assigns Source IP (private) + Source Port
[CGNAT Router] → translates to Translated IP (public) + Translated Port
     ↓
[Public Internet]
     ↓ connects to:
[Destination Server] (WhatsApp server, Telegram server, or another mobile device)
     ↓
[IPDR Logger on Operator Network]
     ↓ records row:
[MSISDN, Source IP, Source Port, Translated IP, Translated Port, Destination IP, Destination Port, Protocol, Duration, Bytes Up, Bytes Down, Cell ID, Start Time, IMEI]
     ↓
```

```
[Telecom Company Database]
     ↓ (upon legal request)
[Exported as CSV/XLSX/Text file]
     ↓
[PRAMAAN IPDR ENGINE — This System]
     ↓ parses, normalises, classifies, and analyses
[Intelligence Dossier / Communication Map / Request Package]
     ↓
[Investigating Officer / Court of Law]
```

---

4. COLUMN NAMES ACROSS INDIAN OPERATORS

The system recognises all of the following source column names and maps them to canonical fields:

MSISDN Field (Suspect's Phone Number):

```text
msisdn, a_party_msisdn, a_number, a_party, subscriber, subscriber_number,
calling_number, calling_party, mobile_number, mdn,
landline_msisdn_mdn_leased_circuit_id,
landline_msisdn_mdn_leased_circuit_id_for_internet_access,
access_identifier
```

Source IP Field (Private/Internal IP):

```text
source_ip, source_ip_address, src_ip, src_address, private_ip,
subscriber_ip, user_ip, allocated_ip, allocated_ip_address,
ip_address_allocated, assigned_ip, source_endpoint, source_ip_port,
source_ip_address_with_source_port
```

Source Port Field:

```text
source_port, source_port_no, source_port_number, src_port, private_port,
subscriber_port, user_port, source_endpoint, source_ip_port,
source_ip_address_with_source_port
```

Translated/NAT IP Field (Public IP):

```text
translated_ip, translated_ip_address, translated_public_ip, public_ip,
public_ip_address, nat_ip, nat_public_ip, mapped_ip, mapped_public_ip,
translated_endpoint, nat_endpoint
```

Translated/NAT Port Field:

```text
translated_port, translated_port_no, translated_port_number, public_port,
nat_port, mapped_port, translated_endpoint, nat_endpoint
```

Destination IP Field (Target Server IP):

```text
destination_ip, destination_ip_address, dest_ip, dst_ip, dst_addr,
server_ip, b_party_ip, b_party_public_ip, remote_ip, remote_address,
peer_ip, destination_endpoint, destination_ip_port,
destination_ip_with_destination_port
```

Destination Port Field:

```text
destination_port, destination_port_no, destination_port_number, dest_port,
dst_port, server_port, b_party_port, remote_port, peer_port,
destination_endpoint, destination_ip_port,
destination_ip_with_destination_port
```

Timestamp Fields:

```text
started_at, start_datetime, start_date_time, timestamp, session_start,
date_time, datetime, ist_start_date+ist_start_time (combined),
start_date_of_ip_address_allocation+start_time_of_ip_address_allocation
```

Duration:

```text
duration_seconds, duration, session_duration, duration_sec, duration_s
```

Upload Bytes:

```text
bytes_up, uplink_bytes, upload_bytes, tx_bytes, bytes_sent, sent_bytes
```

Download Bytes:

```text
bytes_down, downlink_bytes, download_bytes, rx_bytes, bytes_received,
received_bytes
```

IMEI:

```text
imei, imei_number
```

Cell ID:

```text
cell_id, cellid, cell_tower_id, tower_id, cgi, ecgi, lac_cell_id,
enodeb_cell_id
```

PART B — SYSTEM ARCHITECTURE

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5. SYSTEM COMPONENTS

The Pramaan IPDR Engine is a two-part application:

5.1 Backend (Python / FastAPI)

- **Framework:** FastAPI — a high-performance Python web framework.

- **Data Engine:** Polars — an ultra-fast data processing library used for parsing large CSV/Excel files. Handles millions of rows efficiently.
- **Storage:** SQLite database file (`evidence_store.sqlite`) for all session data. JSON file (`evidence_store.json`) for metadata (cases, uploads, settings).
- **Classification Engine:** Custom IP classification pipeline using ASN lookup, platform range matching, and heuristic analysis.
- **Threading:** All heavy ingestion tasks (file parsing, row validation, database insertion) run in FastAPI's background thread pool — the server stays responsive during ingestion.
- **Audit System:** Every operation is logged to `audit_logs` in-memory and persisted.

5.2 Frontend (React / Vite)

- **Framework:** React 18 with Vite build system.
- **Communication Map:** D3.js force-directed layout graph with WebGL-accelerated rendering.
- **Charts:** Recharts library for area charts, bar charts, and timelines.
- **Animations:** Framer Motion for smooth card transitions and staggered panel entrances.
- **GIS Map:** Leaflet.js for geographic tower location visualisation.
- **State Management:** React useState/useEffect with polling for live ingestion progress.

5.3 Communication Protocol

- The frontend communicates with the backend via a REST API at `/api/` .
- API endpoints are documented and accessible at `http://localhost:8000/docs` (FastAPI Swagger UI).

---

6. FILE FORMATS SUPPORTED

FORMAT	EXTENSION	NOTES
Comma-Separated Values	`.csv`	Most common. Auto-detected delimiter.
Tab-Separated Values	`.tsv`	Also .txt with tab delimiter.
Pipe-Separated Values	`.txt`	Detected automatically by delimiter sniffer.
Semicolon-Separated	`.csv` / `.txt`	Used by some European telecom exports.

FORMAT	EXTENSION	NOTES
Microsoft Excel	`.xlsx` / `.xls`	First sheet is parsed. Infer schema turned off to preserve string accuracy.
JSON Array	`.json`	Must be an array of objects or `{"records": [...]}` format.
ZIP Archive	`.zip`	Contains multiple CSV/Excel files. Each member is processed individually and merged.
RAR Archive	`.rar`	**NOT SUPPORTED.** Extract to ZIP or CSV first.

Maximum Upload Size: Configurable. Default is set in the server's configuration.

---

7. COLUMN ALIAS RESOLUTION ENGINE

When a file is uploaded, the system performs the following normalisation pipeline:

1. **Read all column headers** from the file.
2. **Strip whitespace and special characters** from column names.
3. **Convert to lowercase** and replace spaces with underscores.
4. **Match against the alias dictionary** (see Section 4 above). For each canonical field, check if any of the file's columns matches a known alias.
5. **If a combined column is detected** (e.g., `source_ip_port` = `10.5.3.2:54231`), the engine automatically splits it into separate IP and Port fields.
6. **If a date+time split** is detected (e.g., separate `start_date` and `start_time` columns), they are merged into a single `started_at` datetime field.
7. **Report missing required fields** — if any of `msisdn`, `source_ip`, `source_port`, `destination_ip`, `destination_port` cannot be mapped, they are listed in the `missing_required` validation report.

---

8. THE CLASSIFICATION ENGINE

Each ingested session row undergoes automatic classification:

Step 1: Custom Platform Range Check

- The destination IP is checked against all active Platform Ranges in the database.

- If it matches a CIDR block (e.g., `149.154.162.5` is within `149.154.160.0/20` = Telegram), the session is immediately classified as:
  - `classification = "relay"`, `confidence = 0.85`
  - `operator = "Telegram Messenger"`, `asn = "AS62041"`
- If the protocol is UDP and port ≥ 1024 and duration > 10 seconds, upgrade to:
  - `app_hint = "Telegram VoIP Call (Relayed)"`, `voip = True`

Step 2: Known STUN/TURN Port Check

- If destination port is 3478, 3479, 3480, or 5349:
  - `classification = "relay"`, `confidence = 0.70`
  - `app_hint = "STUN/TURN signalling"`

Step 3: P2P VoIP Detection

- If destination IP is NOT a known platform AND bytes_down > 0 AND port ≥ 1024 AND protocol = UDP AND duration > 10 seconds:
  - `classification = "p2p"`, `confidence = 0.90`
  - `app_hint = "Direct P2P VoIP Call"`

Step 4: Unknown

- All other sessions:
  - `classification = "unknown"`, `confidence ≈ 0.48`

---

9. DATABASE AND PERSISTENCE LAYER

The system uses a two-tier storage strategy:

9.1 SQLite Database (`evidence_store.sqlite`)

Stores all session records (high-volume data). Tables:

- **sessions:** Every ingested IPDR row — indexed by `case_id`, `a_party_msisdn`, `destination_ip`, `classification`, and `upload_id` for fast queries.
- **uploads / ingestion_jobs:** Upload history and job progress tracking.
- **investigation_reports:** Extraction results and request packages.
- **audit_logs:** System audit trail.

9.2 JSON Metadata (`evidence_store.json`)

Stores all configuration data (low-volume):

- **cases:** All created cases.
- **import_specs:** Custom column mapping specifications.
- **platform_ranges:** Custom IP classification CIDR blocks.

9.3 Evidence Files (`evidence_files/` directory)

Physical copies of all uploaded raw carrier log files are stored here for chain-of-custody purposes.

9.4 Startup Recovery

On every system restart, the engine:

1. Scans all Jobs in status `"processing"` and marks them `"failed"` with message `"Job interrupted (server restarted)"`.
2. Scans all Uploads in status `"processing"` and marks them `"failed"` with the same message.
3. This prevents ghost jobs from appearing permanently stuck in the UI.

PART C — STEP-BY-STEP OPERATIONS

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10. STEP 1 — SETTING UP CASES

Before uploading any carrier data, you must create a **Case** to organise the investigation.

10.1 What is a Case?

A Case is an investigation container. Every upload, extraction, report, and request package is associated with one case. This provides:

- **Isolation:** Evidence from Case A does not contaminate analysis of Case B.
- **Filtering:** Dashboard charts, communication maps, and reports can be filtered to show only one case's data.
- **Chain of custody:** Case creation is logged in the audit trail with timestamp.

10.2 Creating a Case

1. Navigate to the **Cases** tab in the left sidebar.
2. Click **+ New Case** in the top right.

3 Fill in the case creation form:

- **Case Name:** Use a unique, structured identifier (e.g., `GWL-CYBER-2024-0042-EXTORTION`). Maximum 120 characters.
- **Crime Type:** Select from the dropdown or type a custom value (e.g., `Extortion`, `Murder`, `Drug Trafficking`, `CSAM`, `Cyber Fraud`). Maximum 80 characters.
- **Investigating Officer Name:** The name of the officer responsible for this case. Maximum 100 characters.
- **Description:** Detailed narrative of the case, suspects, and investigation objectives.
- **Target MSISDNs:** Enter one or more suspect phone numbers (one per line). This pre-populates the PoI lookup for quick access later.
- **Tags:** Optional classification tags (e.g., `gang-case`, `financial`, `narcotic`).

4 Click **Save Case**.

10.3 Case Statuses

STATUS	MEANING
`active`	Case is currently under investigation. All operations allowed.
`review`	Case is under legal/supervisory review. Data is read-only.
`closed`	Case closed. Data retained but no modifications allowed.

10.4 Deleting a Case

- The default general case (`CASE-GENERAL`) is **protected** and cannot be deleted.
- Custom cases can be deleted. Deletion removes the case record but does NOT delete the associated session data in SQLite.
- To fully wipe all data, use the Factory Reset function (Section 26).

---

11. STEP 2 — UPLOADING CARRIER LOGS

11.1 Before You Upload

Ensure you have:

1 The IPDR log file in a supported format (CSV, XLSX, ZIP — see Section 6).

2 The correct Case selected.

3 Basic knowledge of which operator produced the file (Jio, Airtel, Vi, BSNL, DoT format).

4. If the format is non-standard, create an Import Specification first (see Section 11.4).

11.2 Upload Procedure

1. Go to the **Uploads** tab in the left sidebar.
2. From the **Case** dropdown at the top, select the relevant case.
3. From the **Import Specification** dropdown:
   - Select `Auto-detect` if you trust the engine's adapter auto-detection.
   - Select a specific operator adapter (`Jio`, `Airtel`, `Vodafone Idea`, `BSNL`, `DoT IPDR`) if you know the source.
   - Select a custom specification you have previously saved.
4. Drag and drop your file onto the upload area, OR click the upload zone to browse and select the file.
5. Click **Upload and Ingest**.

11.3 Understanding the Progress Display

Once uploaded, the ingestion job appears in the **Recent Ingestion Jobs** panel. The status card shows:

- **Status badge:** `processing` (animated, blue), `completed` (green), `failed` (red).
- **Progress bar:** Fills from 5% to 100% as rows are processed.
- **Row counters:** Live counts of Rows Total / Valid / Quarantined.
- **Ingestion speed:** Rows per second throughput.
- **Time elapsed:** Total duration of the ingestion process.

If the server is restarted during ingestion, the job will appear as `failed` (100%) on next startup — this is the zombie recovery mechanism.

11.4 Creating an Import Specification

If the operator's file uses non-standard column names:

1. Go to **Settings > Import Specs**.
2. Click **+ New Import Spec**.
3. Enter a name (e.g., `Airtel Indore - Custom Format`).
4. Use the **Auto-suggest mapping** feature: paste or upload a small sample of the file, and the system will suggest mappings based on column name similarity.

5. Review and adjust each mapping:
   - Left column: The canonical field name (e.g., `msisdn`)
   - Right column: The exact column name from the operator's file (e.g., `calling_party_number`)
6. Set the **delimiter** if the system cannot auto-detect it.
7. Click **Save**. The specification is now available in the upload dropdown.

11.5 Uploading ZIP Archives

ZIP archives containing multiple CSV files are supported. The system:

1. Extracts each member file.
2. Parses each independently.
3. Merges all rows into a single ingestion batch.
4. Tags each row with its source filename (`source_file` field).
5. Reports per-member statistics in the format report.

---

12. STEP 3 — VALIDATING UPLOADS AND QUARANTINE REVIEW

12.1 Reading the Upload Status Card

After ingestion, click the upload record to expand its details. You will see:

- **Format Report:** Parser engine used, file format, delimiter, detected adapter, encoding, and list of all detected columns.
- **Missing Required Fields:** Any of the five required fields (`msisdn`, `source_ip`, `source_port`, `destination_ip`, `destination_port`) that were not found. If any are missing, investigation capability is severely limited.
- **Quarantine Count:** Number of rows rejected due to validation failures.
- **Notes:** Any warnings generated during parsing (e.g., unsupported archive members, encoding issues).

12.2 Reviewing Quarantined Rows

Click **View Quarantine** to see the list of rejected rows:

- **Row Number:** The exact row in the source file that was rejected (useful for manual inspection).
- **Field:** Which specific field caused the failure.
- **Reason:** Human-readable explanation (e.g., `"Invalid IP address: 999.0.0.1"`, `"Port out of range: 99999"`).

12.3 Acceptable Quarantine Rates

QUARANTINE %	ASSESSMENT
0–5%	Excellent. Minimal data quality issues.
5–15%	Good. Some corrupt rows, typical for real-world data.
15–30%	Concerning. Format may be partially mismatched. Review the quarantine log.
>30%	Critical. The Import Specification is likely wrong. Do not proceed with analysis. Re-upload with correct mapping.

---

13. STEP 4 — READING THE DASHBOARD

The Dashboard provides a high-level overview of all ingested evidence across all cases.

13.1 Stat Cards (Top Row)

CARD	WHAT IT SHOWS
Total Cases	Number of investigation cases created in the system
Total Uploads	Number of carrier log files successfully ingested
Total Sessions	Number of individual IPDR session rows in the database
Actionable (P2P)	Number of sessions classified as direct P2P communication events
Platform Relays	Number of sessions classified as application server relay traffic
Unknown / Other	Number of sessions that could not be confidently classified
Avg. Confidence	Mean confidence score across all classified sessions
Quarantined Rows	Total rows rejected across all uploads

13.2 Suspicious Patterns Panel

Lists all auto-detected anomalies. Each pattern has:

- **Severity:** High / Medium / Low (colour coded red/yellow/blue)
- **Pattern Type:** The specific anomaly category.

- **Entities:** The MSISDNs, IMEIs, or IPs involved.
- **Evidence:** Supporting session IDs.
- **Recommended Action:** Suggested next investigative step.

13.3 Latest Upload Panel

Shows the most recently ingested file with its status, row counts, and a link to the full upload detail view.

---

14. STEP 5 — USING THE COMMUNICATION MAP

The Communication Map is a force-directed graph visualisation of all communication flows between nodes (MSISDNs, IP addresses).

14.1 Understanding the Graph

- **Nodes (Circles):** Each node represents either an MSISDN (suspect's phone number) or a Destination IP/service.
  - **Green nodes:** Source MSISDN nodes (suspects' phones).
  - **Red nodes:** P2P destination IP nodes (potential direct call counterparts — high value).
  - **Orange/yellow nodes:** Relay server nodes (application infrastructure).
  - **Grey nodes:** Unknown destination nodes.
- **Edges (Lines connecting nodes):** Each edge represents one or more session flows between the connected nodes.
  - **Edge thickness:** Proportional to the number of sessions between the two nodes.
  - **Edge colour:** Reflects the predominant classification (red = p2p, blue = relay, grey = unknown).
- **Node size:** Proportional to the total byte volume or session count of that node.

14.2 Performance and Density Controls

For large datasets (100K+ sessions), the full graph cannot be rendered without browser lag. The system applies a **Hotspot Density Limiter**:

- Default: Top **250 flows** sorted by session count are displayed.
- Adjustable: Use the **Density Selector** dropdown in the top actions bar to change to `Top 100`, `Top 250`, `Top 500`, or `Top 1000`.
- For datasets with millions of sessions, keep this at `Top 100` for smooth performance.

14.3 Filtering the Graph

Use the actions bar to filter:

- **MSISDN filter:** Enter a specific phone number to display only that node and its direct connections.
- **Classification filter:** `P2P only` (direct calls), `Relay only` (app traffic), `All`.
- **Case filter:** Show flows from one specific case only.
- **Scan limit:** Maximum number of sessions scanned from the database (default 20,000). Increase for more comprehensive graphs on powerful machines.

14.4 Node Detail Card

Click any node to open its detail card on the right side of the screen:

- Node type, label, and operator/ASN.
- Session count and total bytes.
- Average confidence score.
- Last seen timestamp.
- List of individual sessions associated with this node (clickable).

14.5 Exporting the Graph

- **Export JSON:** Downloads the complete graph as a JSON file (nodes + links + sessions). For import into Gephi, Cytoscape, or custom analysis.
- **Export GraphML:** Downloads in GraphML XML format compatible with IBM i2 Analyst's Notebook, yEd, and Gephi.

---

15. STEP 6 — RUNNING B-PARTY EXTRACTIONS

B-Party Extraction is the core analytical function that identifies potential communication counterparts.

15.1 What the Extraction Does

1. Retrieves all sessions for the target MSISDN from the database.
2. Filters sessions by the specified minimum confidence threshold.
3. For each qualifying session, creates an **ExtractionCandidate** record containing:
   - The exact destination IP and port.
   - The translated (public) IP and port.
   - The session timestamp.
   - The classification and confidence.
   - The operator ASN (so you know which carrier to send the CGNAT request to).

4. Packages the candidates into **Request Packages** — pre-formatted legal notice templates.

15.2 Running an Extraction

1. Go to the **B-Party Extraction** tab.
2. Enter the target **MSISDN** in the search field.
3. Set **Depth**:
   - `1` (recommended for initial investigation): Direct connections only.
   - `2` (extended network): Adds one level of association — if IP-A is connected to IP-B, IP-B's known associations are also included.
4. Set **Minimum Confidence** threshold (default `0.65`). Do not go below `0.50`.
5. Click **Run Extraction**.
6. The extraction result shows:
   - **Total sessions analysed** for this MSISDN.
   - **Actionable (P2P) count** and **Relay count**.
   - **Candidate table:** Each candidate B-party IP with its operator, confidence, classification, and evidence session reference.

15.3 Interpreting Candidates

CONFIDENCE	ACTION
≥ 0.90 + P2P + duration > 30s	High priority. Generate Request Package and submit CGNAT search immediately.
0.75–0.89 + P2P	Strong indicator. Generate Request Package. Corroborate with CDR data.
0.65–0.74 + relay	Moderate. Note the application used. May not require CGNAT search unless pattern repeats.
< 0.65	Discard. Background traffic.

---

16. STEP 7 — BUILDING REQUEST PACKAGES

16.1 What a Request Package Contains

A Request Package is a structured data export used to draft legal CGNAT search notices to telecom operators. Each package contains:

```text
Package ID:         PKG-XXXX
Extraction ID:      EXT-XXXX (the extraction this came from)
Session ID:         SESS-XXXX
Request Type:       CGNAT Reverse Lookup
Target Operator:    Reliance Jio Infocomm Ltd. (AS55836)
Payload:
  - Public IP (Translated IP):    103.74.19.155
  - Public Port (Translated Port): 54231
  - Session Start Time (IST):     2024-03-15 14:23:45
  - Session End Time (IST):       2024-03-15 14:25:12
  - Protocol:                     UDP
  - Destination IP:               91.108.56.12
  - Destination Port:             443
```

16.2 Using Request Packages

1. Review the packages in the **Request Packages** section of the B-Party Extraction page.
2. Both columns (left: Package list, right: Package detail) scroll independently, allowing you to compare multiple packages simultaneously.
3. Click **Export** to download the package details as a structured document.
4. Submit the package contents to the identified operator's Law Enforcement Response Team (LERT) via the appropriate legal process.

---

17. STEP 8 — ANALYSING PERSON OF INTEREST (POI) PROFILES

17.1 Accessing a PoI Profile

Method 1: Use the **global search bar** at the top of the app and type the target MSISDN.
Method 2: Click any MSISDN link in the Sessions list.
Method 3: Navigate to **Analytics > PoI Report** and enter the MSISDN.

17.2 PoI Dossier Sections

Summary Stats Row:

STAT	MEANING
Total Sessions	All IPDR records for this MSISDN
Locations Visited	Number of unique Cell ID base stations logged

STAT	MEANING
Applications Used	Number of distinct application platforms detected
Handsets (IMEIs)	Number of distinct physical devices used

Top Locations Panel:

- Lists the top 5 Cell IDs this MSISDN connected from, sorted by visit count.
- Each Cell ID can be cross-referenced with the operator's cell tower database to obtain GPS coordinates.

Associated IMEIs Panel:

- Lists all IMEI numbers logged with this MSISDN.
- Flags any IMEI that also appears with other MSISDNs (Shared Handset alert).

Top Applications Grid:

- Lists all detected application platforms (WhatsApp, Telegram, Signal, Google, etc.) with session counts and cumulative duration in seconds.

Top 10 Application Duration Bar Graph:

- Visual horizontal bar graph showing relative time spent on each application.
- Useful for quickly identifying the primary communication platform used.

WhatsApp Interactions:

- Specifically isolates sessions that match Meta/WhatsApp IP ranges.
- Shows target IP, timestamp, and classification.
- The count badge shows total WhatsApp relay sessions.

17.3 Exporting the PoI Dossier

Click **Export PDF Brief** at the top of the dossier to download a print-ready PDF report.
The PDF contains:

- MSISDN and summary statistics.
- Top 50 sessions table (timestamp, destination IP, port, classification, operator).

18. STEP 9 — USING THE IMEI INTELLIGENCE MODULE

18.1 Accessing IMEI Intelligence

Navigate to **Analytics > IMEI Intelligence** in the sidebar.

18.2 What It Shows

- **Top 10 Most Active Handsets (Bar Chart):** Visual comparison of session counts and unique MSISDNs per IMEI.
- **Global IMEI Frequency Table:** Every unique IMEI found across all uploaded evidence, with:
  - IMEI number
  - Detected handset model (TAC-decoded)
  - Total session count
  - **Unique Suspects (MSISDNs):** Number of different phone numbers this device has been used with. If > 1, it's a **CRITICAL ALERT** (shared handset).

18.3 Shared Handset Alert

When an IMEI appears with multiple MSISDNs:

- The cell shows a **Red badge** with count and the text "Shared Handset!"
- Below it, all MSISDNs associated with this device are listed.

Investigative Response:

- Check the timestamps: Did the multiple MSISDNs use this IMEI simultaneously (impossible — indicates data error) or sequentially (normal SIM swap)?
- Sequential use of the same IMEI across multiple MSISDNs is strong evidence of:
  - A criminal using multiple SIM cards on one phone.
  - A gang sharing one device.
  - Burner SIM strategy to evade tracking.

19. STEP 10 — USING THE LOCATION INTELLIGENCE MODULE

19.1 Accessing Location Intelligence

Navigate to **Analytics > Location Intelligence** in the sidebar.

19.2 What It Shows

A summary table of all Cell IDs with:

- **Cell ID / Label:** The tower identifier.
- **Total Sessions:** Number of sessions logged from this tower.
- **Day Sessions vs. Night Sessions:** Sessions split by IST daytime (06:00–22:00) vs. nighttime (22:00–06:00).
- **Unique MSISDNs:** Number of different suspects seen connecting from this tower.
- **First Seen / Last Seen:** Temporal range of activity at this tower.
- **Coordinates:** Estimated latitude/longitude (from the cell decoder — note these are approximate and centered on the tower area, not exact device GPS).

19.3 Forensic Use Cases

- **Primary Location:** The Cell ID with the highest sessions for a specific MSISDN is likely the suspect's home or workplace.
- **Night Presence:** Sessions predominantly between 22:00–06:00 at one cell ID indicate the suspect's overnight location (home or hideout).
- **Movement Pattern:** Session timestamps across multiple Cell IDs can reconstruct a movement timeline.

---

20. STEP 11 — USING THE ANALYTICS AND TIMELINE

20.1 Timeline View

Navigate to **Analytics > Timeline**.

The timeline shows session activity bucketed by time interval:

- **Bucket options:** Year, Month, Day, Hour, Minute, Second.
- **Y-axis:** Session count.
- **Colour-coded stacked areas:** P2P sessions (green), Relay sessions (red/orange), Unknown (grey).

How to read it:

- A spike in P2P sessions at a specific hour indicates heightened direct communication at that time.
- A spike in relay sessions indicates application usage at that time.
- Sustained high P2P activity over multiple hours indicates a prolonged communication event (e.g., coordinated operation).

The timeline scrolls independently — on large datasets you can scroll horizontally through time periods while keeping the Y-axis labels visible.

20.2 Application Summary View

Navigate to **Analytics > Applications**.

Shows all detected applications across all cases:

- **Name:** Application platform (e.g., "Telegram Messenger").
- **Operator/ASN:** The network owner of the destination IPs.
- **Sessions:** Total session count.
- **Unique MSISDNs:** Number of different suspects using this application.
- **Distinct Destination IPs:** Number of unique servers accessed (higher count may indicate broader coordination or VPN usage).
- **Duration:** Total cumulative seconds of sessions.
- **Bytes Total:** Total data transferred.

20.3 Suspicious Patterns View

Navigate to **Analytics > Suspicious Patterns**.

Lists all auto-detected anomalies with:

- Severity badge (High/Medium/Low).
- Pattern description.
- Involved entities (MSISDNs, IMEIs, IPs).
- Evidence session references.
- Recommended action.

---

21. STEP 12 — USING THE GIS MAP

21.1 Accessing the GIS Map

Navigate to **GIS Map** in the left sidebar.

21.2 What the GIS Map Shows

The GIS Map plots all known cell tower locations (from ingested Cell ID data) as heat map points on an interactive street map (OpenStreetMap tiles).

- **Green circles:** Cell towers with moderate activity.

- **Red/hot zones:** Cell towers with the highest session counts (hotspots).
- **Marker clusters:** In zoomed-out view, nearby towers are clustered into count badges.

21.3 Performance Controls

For datasets with 100K+ sessions, the GIS map applies the **Hotspot Density Limiter**:

- Default: **Top 250 hotspot cell towers** by session count are displayed.
- Adjustable via the density control in the map toolbar.
- **Do not set this above 500 on standard hardware** — more nodes cause significant browser slowdown.

21.4 Interacting with the Map

- **Scroll/pinch:** Zoom in or out.
- **Click marker:** Opens a popup showing the Cell ID, estimated operator, session count, day vs. night sessions, and associated MSISDNs.
- **Click cluster:** Zooms in to reveal individual tower markers.

---

22. STEP 13 — IP INTELLIGENCE REPORTS

22.1 Accessing IP Reports

From any session list, click on a destination IP address to navigate to its Intelligence Report.

22.2 What an IP Report Shows

- **Destination IP:** The IP being investigated.
- **Classification:** How the system classified connections to this IP (p2p / relay / unknown).
- **Operator/ASN:** Who owns this IP block.
- **Total Sessions:** How many sessions across all MSISDNs contacted this IP.
- **Associated MSISDNs:** List of all suspects whose IPDR shows connections to this IP.
- **Port Distribution:** All ports accessed on this IP.
- **First / Last Seen:** Temporal range of activity.
- **Total Bytes:** Data volume.

22.3 Export

- **Export CSV:** Download the full session list for this IP as CSV.
- **Export HTML:** Download a formatted HTML report for court submission.

23. STEP 14 — EXPORTING EVIDENCE

23.1 Export Options Summary

EXPORT TYPE	LOCATION	FORMAT	CONTENTS
PoI Report CSV	PoI Dossier page	`.csv`	Session list for one MSISDN
PoI Report HTML	PoI Dossier page	`.html`	Formatted forensic brief
PoI Report PDF	PoI Dossier page	`.pdf`	Print-ready court brief
IP Report CSV	IP Report page	`.csv`	Sessions for one destination IP
IP Report HTML	IP Report page	`.html`	Formatted IP analysis report
Sessions CSV (all)	Analytics page	`.csv`	All sessions across all cases
Sessions XLSX	Analytics page	`.xlsx`	Multi-sheet Excel workbook
Graph JSON	Communication Map	`.json`	D3-compatible graph structure
Graph GraphML	Communication Map	`.graphml`	IBM i2 / Gephi compatible

23.2 Section 65B Compliance

For all electronic evidence submitted in Indian courts under Section 65B of the Indian Evidence Act, 1872:

- The IO must provide a signed Section 65B certificate alongside any export.
- The certificate must state: the computer system was functional, the evidence was produced in the ordinary course of activities, and the record has not been altered.
- The Audit Log from the system serves as supporting documentation of the chain of custody.

24. STEP 15 — SYSTEM SETTINGS AND PLATFORM RANGES

24.1 Accessing Settings

Navigate to **Settings** in the left sidebar.

24.2 Platform Ranges Management

Platform Ranges are the core of the classification engine. Keeping them accurate is critical.

Adding a New Platform Range:

1. Click + **Add Range**.
2. Enter:
   - **Platform Name:** Human-readable name (e.g., `Truecaller`).
   - **CIDR:** The IP address block (e.g., `40.76.0.0/14`).
   - **ASN:** The autonomous system number (e.g., `AS8075`).
   - **Description:** Brief description (e.g., `Microsoft Azure — used by Truecaller servers`).
   - **Active:** Toggle on/off without deleting the range.
3. Click **Save**.

Deactivating a Range:

- Toggle the **Active** switch off. The range remains in the database but is not used for classification until re-activated.
- All future ingested sessions will not match this range while deactivated.
- Previously classified sessions are NOT retroactively reclassified when you change platform ranges.

Verifying Platform Ranges:
Platform companies change their IP allocations regularly. Cross-reference with:

- RIPE NCC, ARIN, or APNIC WHOIS lookup tools.
- Meta's published IP ranges: https://developers.facebook.com/docs/messenger-platform/
- Google's published IP ranges: https://www.gstatic.com/ipranges/goog.json

24.3 Import Specifications

See Section 11.4 for the full guide on creating Import Specifications.

24.4 System Information Panel

Shows:

- Backend version and storage paths.
- Total counts: Cases, Uploads, Sessions, Reports.
- Last persistence snapshot timestamp.
- **Write Snapshot:** Manually triggers a database backup snapshot to disk.

25. STEP 16 — AUDIT LOGS

25.1 Accessing the Audit Log

Navigate to **Settings > Audit Logs** or **System > Audit Trail**.

25.2 Reading the Audit Log

Each entry shows:

- **Timestamp:** IST datetime of the action.
- **Action:** The operation performed (e.g., `upload_ingested`, `extraction_created`, `platform_range_added`, `reset_executed`).
- **Entity Type:** The category of object affected (e.g., `upload`, `extraction`, `platform_range`).
- **Entity ID:** The specific record affected.
- **Details:** JSON payload with additional context.

25.3 Audit Actions Reference

ACTION	TRIGGER
`upload_ingested`	File successfully uploaded and ingested
`upload_failed`	File upload failed validation
`extraction_created`	B-Party extraction was run
`package_generated`	Request Package was created
`case_created`	New case was created
`case_deleted`	Case was deleted
`platform_range_added`	New CIDR range added to platform settings
`platform_range_deleted`	CIDR range removed
`sqlite_snapshot`	Manual or automatic database snapshot created
`reset_executed`	Factory reset was performed — **critical event**

25.4 Legal Importance

The audit log must be preserved unmodified for the duration of any criminal proceeding in which this system's output is used as evidence. Export a copy of the audit log as part of every case closure process.

26. STEP 17 — FACTORY RESET / DANGER ZONE

26.1 What the Factory Reset Does

The Factory Reset permanently and irreversibly:

1. Deletes all session data from the SQLite database (`evidence_store.sqlite`).
2. Deletes all uploaded evidence files from the `evidence_files/` directory.
3. Resets all case records, upload histories, extraction results, and request packages to zero.
4. Executes `VACUUM` on the SQLite database to reclaim disk space.
5. Retains only the system configuration (platform ranges, import specs) — these survive a reset.

26.2 When to Use It

- At the conclusion of a case when the workstation must be cleared of all investigation data before deployment to a new case or field location.
- When test/synthetic data was ingested during system setup and needs to be cleared before real-case data is loaded.
- Upon explicit orders from supervisory authority.

26.3 Performing a Factory Reset

1. Navigate to **Settings > System Info**.
2. Scroll to the **Danger Zone** panel (red section at the bottom).
3. Click **Reset to 0**.
4. A confirmation dialog will appear with a warning message.
5. Read the warning carefully.
6. Click **Confirm Reset**.
7. The system immediately wipes all data and returns to a clean state.

THIS ACTION CANNOT BE UNDONE. THERE IS NO RECYCLE BIN OR RECOVERY. ENSURE ALL EVIDENCE HAS BEEN EXPORTED BEFORE PROCEEDING.

PART D — FORENSIC VERIFICATION CHECKLIST

27. PRE-REPORT VALIDATION CHECKLIST

Before submitting any IPDR analysis report to a court, superior officer, or field operation team, verify every item on this checklist:

```text
[ ] 1. TIMEZONE VERIFICATION
        All session timestamps confirmed as IST (UTC+05:30)?
        If source logs were in UTC: have you added 5 hours 30 minutes?
        If logs were in UTC+0: sessions at e.g., 08:00 UTC = 13:30 IST.

[ ] 2. CGNAT PORT INCLUDED IN REQUEST PACKAGES
        Every Request Package includes BOTH translated IP and translated port?
        Timestamp in Request Package is exact to the second?

[ ] 3. CONFIDENCE THRESHOLD RESPECTED
        No session below confidence 0.65 used as standalone evidence?
        Any session below 0.80 cross-referenced with CDR or physical surveillance?

[ ] 4. QUARANTINE RATE ACCEPTABLE
        Quarantine rate for all uploads below 30%?
        All quarantined rows reviewed and their exclusion justified?

[ ] 5. SESSION DURATION VERIFIED
        All P2P sessions used as evidence have duration > 10 seconds?
        Sessions with duration < 5 seconds explicitly excluded from call evidence?

[ ] 6. IMEI CROSS-REFERENCE
        Any target MSISDN with zero sessions checked for IMEI overlap with active
MSISDNs?
        All Shared Handset alerts reviewed and documented?

[ ] 7. PLATFORM RANGES UP TO DATE
        Platform ranges verified against current operator IP allocations?
        No IP reclassification changes made after evidence was analysed (retroactive
        reclassification is not applied — all classification is fixed at ingestion
time)?

[ ] 8. AUDIT LOG PRESERVED
        Complete audit log exported and signed?
        No entries deleted from audit log during investigation?

[ ] 9. CASE CORRECTLY ASSIGNED
        All uploaded evidence correctly associated with the correct case ID?
        No cross-contamination from other cases?

[ ] 10. EXPORT METHOD
         Evidence exported as PDF/CSV with Section 65B certificate prepared?
```

```
        Hash (SHA-256) of exported files computed and documented for integrity
verification?
```

---

28. COMMON MISTAKES AND HOW TO AVOID THEM

Mistake 1: Using IP Address Alone (Without Port) for CGNAT Lookup

What happens: You send a CGNAT search request to an operator with only an IP address and no port. The operator returns "no match" or multiple matches.
Why: Thousands of subscribers share one public IP at any given second. Only the port uniquely identifies one subscriber.
Correct approach: Always include translated_ip + translated_port + exact timestamp.

Mistake 2: Confusing Source IP with Translated IP

What happens: You submit the subscriber's private `10.x.x.x` IP to an external platform's law enforcement portal requesting subscriber identity.
Why it fails: Private IPs are internal to the carrier network and are not visible on the public internet.
Correct approach: Use `translated_ip` (the public NAT IP) for any external platform searches.

Mistake 3: Treating Every WhatsApp Relay Session as a "WhatsApp Call"

What happens: You report that the suspect made 500 WhatsApp calls based on 500 relay sessions.
Why it is wrong: WhatsApp relay sessions include text messages, media downloads, status syncs, and notification checks — NOT only calls.
Correct approach: Filter by UDP protocol + duration > 10 seconds to identify actual call events.

Mistake 4: Ignoring Timezone in Exported Logs

What happens: Operator exports log times in UTC but labels them without timezone. Analyst assumes IST.
Why it is wrong: At 14:00 UTC, IST is 19:30. An alibi for 14:00 local time is irrelevant if the session was at 14:00 UTC = 19:30 IST.
Correct approach: Confirm the timezone with the operator's LERT team before analysis.

Mistake 5: Using Confidence < 0.65 in Court Documents

What happens: A session with confidence 0.48 (unknown classification) is cited as evidence of a WhatsApp call.
Why it is wrong: 0.48 confidence means the system could not reliably classify this session. It is statistically noise.
Correct approach: Use only sessions with confidence ≥ 0.65 (for corroborative evidence) or ≥ 0.80 (for primary evidence).

Mistake 6: High Quarantine Rate Without Investigation

What happens: 40% of rows in a carrier's log are quarantined. The analyst proceeds with the remaining 60%.
Why it is wrong: The quarantined 40% may contain the most relevant sessions. The quarantine was caused by a format mismatch, not data corruption.
Correct approach: Review the quarantine log, identify the error pattern, create a correct Import Specification, and re-upload the file.

Mistake 7: Factory Reset During Active Investigation

What happens: The system is reset while a case is still under investigation, wiping all evidence.
Why it happens: The reset function is in Settings > System Info and may be inadvertently triggered.
Prevention: Ensure all evidence is exported (CSV/PDF) and stored on a secure, backed-up server before resetting. Only authorised senior officers should have access to the Settings section.

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29. LEGAL ADMISSIBILITY NOTES

29.1 Governing Law

IPDR analysis and its use as digital evidence in Indian courts is governed by:

- **Indian Evidence Act, 1872** — Section 65B (electronic records admissibility)
- **Information Technology Act, 2000** — Sections 43A (data protection), 69 (interception authorisation)
- **Code of Criminal Procedure (CrPC), 1973** — Sections 91, 92 (production of documents), Section 164 (statements)
- **Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023** — Sections 94, 95 (replacing CrPC 91/92)
- **Telecom Cyber Security Rules, 2024**
- **K.S. Puttaswamy v. Union of India (2017)** — Constitutional right to privacy

29.2 Requirements for Section 65B Certificate

The Section 65B certificate must be provided by a person in a responsible official position who:

1. Has lawful control over the computer system from which the electronic record was produced.
2. Certifies that the computer system was in proper working condition at the relevant time.
3. Certifies that the information was produced in the ordinary course of activities.
4. Certifies that the electronic record has not been altered.

The Audit Log from this system serves as supporting documentation for points 1, 3, and 4 above.

29.3 Peer Review Requirement

No IPDR analysis report should be submitted as court evidence without independent review by at least one other qualified digital forensics examiner. The system's confidence scores and classifications are probabilistic tools, not definitive legal findings.

PART E — QUICK REFERENCE

30. PORT-TO-APPLICATION QUICK REFERENCE TABLE

PORT	PROTOCOL	APPLICATION/SERVICE
80	TCP	HTTP (unencrypted web)
443	TCP	HTTPS (encrypted web, WhatsApp messages, Telegram text)
443	UDP	QUIC/HTTP3 (WhatsApp calls, Google Meet, YouTube)
853	TCP/UDP	DNS over TLS
3478	UDP	STUN (VoIP call setup — WhatsApp, Zoom, Signal)
3479	UDP	STUN alternate
5349	TCP/UDP	STUN over TLS
5222	TCP	XMPP (Jabber-based messaging)
5269	TCP	XMPP server-to-server
5228	TCP	Google Cloud Messaging (Android push notifications)
4244	TCP	Telegram MTProto (primary)
4242	TCP	Telegram MTProto alternate
19302	UDP	Google STUN (WebRTC / Google Meet)
19305	UDP	Google TURN (WebRTC relay)
10000–65535	UDP	RTP/SRTP media streams (voice/video calls — dynamic)
8080	TCP	HTTP alternate / proxy
8443	TCP	HTTPS alternate

PORT	PROTOCOL	APPLICATION/SERVICE
1194	UDP	OpenVPN
1723	TCP	PPTP VPN
500	UDP	IKEv2 VPN (IPSec)
4500	UDP	IKEv2 VPN NAT traversal
51820	UDP	WireGuard VPN

31. OPERATOR IPDR COLUMN NAME REFERENCE TABLE

Jio IPDR Column Names

CANONICAL FIELD	JIO COLUMN NAME
msisdn	msisdn
source_ip	src_ip
source_port	src_port
translated_ip	translated_ip
translated_port	translated_port
destination_ip	destination_ip
destination_port	destination_port
duration_seconds	duration
bytes_up	uplink_bytes
bytes_down	downlink_bytes
started_at	timestamp
imei	imei

Airtel IPDR Column Names

CANONICAL FIELD	AIRTEL COLUMN NAME
msisdn	a_party_msisdn
source_ip	source_ip
source_port	source_port
destination_ip	dest_ip
destination_port	dest_port
duration_seconds	session_duration
bytes_up	uplink_bytes
bytes_down	downlink_bytes
started_at	start_date_time

DoT IPDR (Standard Format) Column Names

CANONICAL FIELD	DOT COLUMN NAME
msisdn	landline_msisdn_mdn_leased_circuit_id_for_internet_access
source_ip	source_ip_address
source_port	source_port
translated_ip	translated_ip_address
translated_port	translated_port
destination_ip	destination_ip
destination_port	destination_port
ip_allocation	static_dynamic_ip_address_allocation
started_at	ist_start_date + ist_start_time (merged)
ended_at	ist_end_date + ist_end_time (merged)

CANONICAL FIELD	DOT COLUMN NAME
imei	imei_number
imsi	imsi
cell_id	cell_id

Vodafone Idea Column Names

CANONICAL FIELD	VI COLUMN NAME
msisdn	subscriber
source_ip	private_ip
source_port	private_port
destination_ip	remote_ip
destination_port	remote_port
bytes_up	tx_bytes
bytes_down	rx_bytes

BSNL Column Names

CANONICAL FIELD	BSNL COLUMN NAME
msisdn	mobile_number
source_ip	allocated_ip
source_port	source_port
destination_ip	server_ip
destination_port	server_port
duration_seconds	duration_sec

32. IP ADDRESS CLASSIFICATION REFERENCE

IP RANGE	TYPE	MEANING
10.0.0.0/8	Private (RFC 1918)	Operator internal network IP (subscriber private address)
172.16.0.0/12	Private (RFC 1918)	Operator internal network IP
192.168.0.0/16	Private (RFC 1918)	LAN / home router IP
100.64.0.0/10	CGNAT (RFC 6598)	Carrier-Grade NAT address space (Jio uses this extensively)
127.0.0.0/8	Loopback	Device's own loopback address. Should never appear in IPDR.
0.0.0.0/8	Reserved	Invalid source address. Quarantined.
255.255.255.255	Broadcast	Network broadcast. Should never appear in IPDR.
224.0.0.0/4	Multicast	Network multicast group. Rare in IPDR.
157.240.0.0/16	Public (Meta)	WhatsApp / Facebook servers
69.171.224.0/19	Public (Meta)	WhatsApp / Facebook servers
31.13.64.0/18	Public (Meta)	WhatsApp / Facebook CDN
149.154.160.0/20	Public (Telegram)	Telegram core servers
91.108.4.0/22	Public (Telegram)	Telegram servers
91.105.192.0/23	Public (Telegram)	Telegram servers
17.0.0.0/8	Public (Apple)	iMessage, FaceTime, Apple servers
142.250.0.0/15	Public (Google)	Google Meet, YouTube, Gmail
74.125.0.0/16	Public (Google)	Google Duo / Meet
52.112.0.0/14	Public (Microsoft)	Microsoft Teams / Skype
66.22.196.0/22	Public (Discord)	Discord VoIP servers
141.101.0.0/16	Public (Cloudflare)	Cloudflare CDN / WARP VPN
1.1.1.1/32	Public (Cloudflare)	Cloudflare DNS (also WARP VPN entry point)

IP RANGE	TYPE	MEANING
8.8.8.8/32	Public (Google)	Google Public DNS

End of PRAMAAN IPDR Intelligence & Correlation Engine User Manual
Document Version 2.0.0 — Full Forensic Edition
Gwalior Police Cyber Forensics Unit

REMINDER: The output of this system is a forensic intelligence tool, not a verdict. Every suspect identified by this system retains the fundamental right to life and personal liberty guaranteed under Article 21 of the Constitution of India. Every finding must be corroborated, every certificate must be signed, and every submission to a court of law must be made in full compliance with the Indian Evidence Act, 1872.